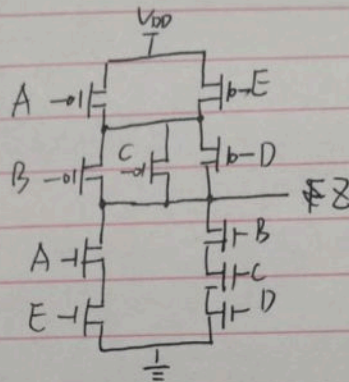


数大第一次作业

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$$\begin{aligned} (-) Z &= \overline{AE} \cdot \overline{BCD} \\ &= \overline{AE + BCD} \end{aligned}$$



(二) ① 对于 nmos 来说:

$$V_{DSatn} = \frac{V_{G1n}}{1 + V_{G1n}/E_{oc}L} = \frac{1.1 - 0.343}{1 + (1.1 - 0.343)/(0.288)} = 0.209 \text{ V}$$

$$\begin{aligned} I_{DSatn} &= V_{Satn} C_{ox} W (V_{G1n} - V_{DSatn}) = V_{Satn} \frac{C_{ox}}{t_{ox}} W (V_{GS} - V_{Tn} - V_{DSatn}) \\ &= 1.09 \times 10^{-5} \cdot \frac{3.5 \times 10^{-11}}{2.37 \times 10^{-9}} \times 101 \times 10^{-9} \times (1.1 - 0.343 - 0.209) = 89.2 \mu\text{A} \end{aligned}$$

② 对于 pmos 来说

$$V_{DSatp} = \frac{V_{G1p}}{1 + V_{G1p}/E_{oc}L} = \frac{V_{G1p}}{1 + (V_{G1p})/(2V_{Satp} \cdot L)} = 0.444 \text{ V}$$

$$I_{DSatp} = V_{Satp} C_{ox} W (V_{G1p} - V_{DSatp}) = 63.6 \mu\text{A}$$

$$\textcircled{3} \frac{I_{DSatn}}{I_{DSatp}} = \frac{89.2}{63.6} = \frac{1.4}{1}$$

造成明显偏差的原因是短沟器件会存在速度饱和效应, 并且空穴的临界电场大于电子的临界电场

(三) 11) $V_{DSAT} = \frac{V_{GT}}{1 + V_{GT}/(E_{oc}L)}$, $\therefore E_{oc}L = 0.906V$, $V_{GT} \gg E_{oc}L$ 不成立
假设 M_1 管在电阻区, 则

$$I_D = \frac{V_{DD} - V_{DS}}{R} = \frac{k'}{1 + V_{DS}/(E_{oc}L)} \cdot \frac{W}{L} \left(V_{GT} + V_{DS} - \frac{V_{DS}^2}{2} \right)$$

代入已知数据, 解得 $V_{DS} = 0.453V < V_{DSAT}$, 所以假设成立
 $\therefore V_{out} = 0.453V$, M_1 处于电阻区

$$\begin{aligned} 12) Q_{before} &= C_{ACB, cutoff} \cdot V_{AB1} + C_{AGS, cutoff} \cdot V_{GS1} + C_{AGD, cutoff} \cdot V_{AD1} \\ &\quad + C_{ASD} \cdot V_{AS1} + C_{APD} \cdot V_{AD1} \\ &= 0 + 0 + 0 + 0 + 0.31 \text{ fF}/\mu\text{m} \times 1 \mu\text{m} \times (0 - 2.5V) = -0.775 \text{ fC} \end{aligned}$$

$$\begin{aligned} Q_{after} &= C_{ACB, triode} \cdot V_{AB2} + C_{AGS, triode} \cdot V_{GS2} + C_{AGD, triode} \cdot V_{AD2} \\ &\quad + C_{ASD} \cdot V_{AS2} + C_{APD} \cdot V_{AD2} \\ &= 0 + \frac{1}{2} \times (6 \text{ fF}/\mu\text{m}^2 \times 1 \mu\text{m} \times 0.25 \mu\text{m}) \times (2.5 - 0) \\ &\quad + \frac{1}{2} \times (6 \text{ fF}/\mu\text{m}^2 \times 1 \mu\text{m} \times 0.25 \mu\text{m}) \times (2.5V - 0.453V) \\ &\quad + 0.31 \text{ fF}/\mu\text{m} \times 1 \mu\text{m} \times (2.5V - 0) + 0.31 \text{ fF}/\mu\text{m} \times 1 \mu\text{m} \times (2.5V - 0.453V) \\ &= 4.82 \text{ fC} \end{aligned}$$

$$\Delta Q = Q_{after} - Q_{before} = 4.82 \text{ fC} - (-0.775 \text{ fC}) = 5.595 \text{ fC}$$